

Logistic Regression

Exam-Level Solved Example | AI / ML Course

Note: In this exam, weights are provided — you do NOT need to apply Gradient Descent. Simply use the given weights to compute predictions and loss.

Given Information

x1	x2	y (Label)
1	2	1
3	1	0
2	3	1

Given Weights: $w_1 = 0.5$ $w_2 = -0.8$ b (bias) = 0.1 | Threshold = 0.5

Step 1 — Compute z (Linear Combination)

Formula:

$$z = (w_1 * x_1) + (w_2 * x_2) + b$$

Sample	Calculation	z
Sample 1	$(0.5 \times 1) + (-0.8 \times 2) + 0.1 = 0.5 - 1.6 + 0.1$	-1.0
Sample 2	$(0.5 \times 3) + (-0.8 \times 1) + 0.1 = 1.5 - 0.8 + 0.1$	0.8
Sample 3	$(0.5 \times 2) + (-0.8 \times 3) + 0.1 = 1.0 - 2.4 + 0.1$	-1.3

Step 2 — Apply Sigmoid Function

Formula:

$$\text{sigmoid}(z) = 1 / (1 + e^{(-z)})$$

Sample	z	Calculation	sigmoid(z)
Sample 1	-1.0	$1 / (1 + e^{1.0}) = 1 / (1 + 2.718) = 1 / 3.718$	0.269
Sample 2	0.8	$1 / (1 + e^{-0.8}) = 1 / (1 + 0.449) = 1 / 1.449$	0.690
Sample 3	-1.3	$1 / (1 + e^{1.3}) = 1 / (1 + 3.669) = 1 / 4.669$	0.214

Step 3 — Classification (Threshold = 0.5)

If $\text{sigmoid}(z) \geq 0.5 \rightarrow$ Predict Class 1 | If $\text{sigmoid}(z) < 0.5 \rightarrow$ Predict Class 0

Sample	$\text{sigmoid}(z)$	Predicted	Actual y	Correct?
Sample 1	0.269	0	1	WRONG
Sample 2	0.690	1	0	WRONG
Sample 3	0.214	0	1	WRONG

All predictions are wrong because the weights are not yet trained — this is expected.

Step 4 — Binary Cross-Entropy Loss

Formula (per sample):

$$L = -[y * \log(y_{\text{hat}}) + (1 - y) * \log(1 - y_{\text{hat}})]$$

Sample 1 ($y=1, y_{\text{hat}}=0.269$)

$$L1 = -[1 * \log(0.269) + 0] = -\log(0.269) = -(-1.312) = 1.312$$

Sample 2 ($y=0, y_{\text{hat}}=0.690$)

$$L2 = -[0 + 1 * \log(1 - 0.690)] = -\log(0.310) = -(-1.171) = 1.171$$

Sample 3 ($y=1, y_{\text{hat}}=0.214$)

$$L3 = -[1 * \log(0.214) + 0] = -\log(0.214) = -(-1.541) = 1.541$$

$$\text{Average Loss: } L_{\text{avg}} = (1.312 + 1.171 + 1.541) / 3 = 4.024 / 3 = 1.341$$

Final Summary Table

Sample	z	$\text{sigmoid}(z)$	Predicted	Actual	Loss
Sample 1	-1.0	0.269	0	1	1.312
Sample 2	0.8	0.690	1	0	1.171
Sample 3	-1.3	0.214	0	1	1.541
				Avg Loss	1.341

Key Formulas to Remember

Step	Formula
Linear Combination	$z = (w1 * x1) + (w2 * x2) + b$

Sigmoid Function	$\text{sigmoid}(z) = 1 / (1 + e^{(-z)})$
Classification	$y_{\text{hat}} = 1$ if $\text{sigmoid}(z) \geq 0.5$, else $y_{\text{hat}} = 0$
Loss (1 sample)	$L = -[y \cdot \log(y_{\text{hat}}) + (1-y) \cdot \log(1-y_{\text{hat}})]$
Average Loss	$L_{\text{avg}} = (1/n) * \text{sum of all } L$

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